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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

CC3002-1 – NEXT GENERATION TELECOM NETWORKS

(For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is a key feature of Heterogeneous Networks (HetNets) in LTE and Beyond 4G?**
A) Uniform coverage using a single cell type B) Integration of macro cells, micro cells, and small cells
C) Exclusive use of high-frequency bands D) Simplification of network infrastructure
2. **Which of the following is NOT considered a key pillar of 5G technology?**
A) Enhanced Mobile Broadband (eMBB) B) Ultra-Reliable Low Latency Communications (URLLC)
C) Massive Machine Type Communications (mMTC) D) High-Frequency Global Positioning System (HF-GPS)
3. **Which pillar of 5G focuses on supporting a massive number of Internet of Things (IoT) devices?**
A) Enhanced Mobile Broadband (eMBB) B) Ultra-Reliable Low Latency Communications (URLLC)
C) Massive Machine Type Communications (mMTC) D) High-Speed Data Transmission (HSDT)
4. **What is the expected latency reduction goal for 5G networks?**
A) From 50 milliseconds to 10 milliseconds B) From 20 milliseconds to 5 milliseconds
C) From 10 milliseconds to 1 millisecond D) From 5 milliseconds to 0.1 millisecond
5. **Which frequency bands are primarily used in 5G for enhanced data speeds?**
A) Low-frequency bands (sub-1 GHz) B) Mid-frequency bands (1-6 GHz)
C) High-frequency bands (30-300 GHz) D) Ultra-low frequency bands (below 100 MHz)
6. **What is a primary challenge associated with the deployment of millimeter-wave (mmWave) frequencies in 5G networks?**
A) Poor penetration through obstacles like buildings and trees B) High latency
C) Limited data rates D) Low power consumption
7. **Which environmental factor can significantly affect the performance of 5G mmWave frequencies?**
A) Temperature variations B) Rain and humidity
C) Solar radiation D) Soil composition
8. **What is the main advantage of beamforming in millimeter-wave communications?**
A) Increased spectral efficiency B) Reduced interference
C) Longer communication range D) Lower power consumption

9. Which architecture component in 5G is responsible for supporting multiple radio access technologies (RATs)?
- A) User Equipment (UE) B) gNodeB (gNB)
C) Multi-RAT Controller (MRC) D) 5G Core (5GC)
10. What is the primary purpose of Software Defined Networking (SDN) in 5G networks?
- A) To provide centralized network control and management B) To increase hardware complexity
C) To reduce data transmission speeds D) To enhance voice call quality
11. What is the role of the SDN controller in a 5G network?
- A) Managing network devices and making routing decisions B) Forwarding packets based on predefined rules
C) Generating user data D) Monitoring signal strength
12. How does SDN contribute to the concept of Network Slicing in 5G?
- A) By dynamically allocating network resources to different slices based on demand B) By reducing the number of network slices
C) By improving hardware efficiency D) By allocating static resources to each network slice
13. What is a primary benefit of D2D communication in 4G and 5G networks?
- A) Increased network latency B) Enhanced spectral efficiency and reduced latency by allowing devices to communicate directly
C) Increased power consumption D) Improved voice call quality
14. In the context of D2D communication, what does proximity service (ProSe) refer to?
- A) Services provided over long distances B) Services that enable direct communication between devices in close proximity
C) Services requiring high latency D) Services that rely solely on the core network
15. Which 5G use case benefits significantly from D2D communication?
- A) Enhanced Mobile Broadband (eMBB) B) Fixed Wireless Access (FWA)
C) Massive Machine Type Communications (mMTC) D) Ultra-Reliable Low Latency Communications (URLLC)
16. How does D2D communication enhance the performance of Internet of Things (IoT) devices in 5G?
- A) By reducing the number of IoT devices B) By increasing the power consumption of IoT devices
C) By enabling direct and efficient communication between IoT devices D) By limiting IoT device connectivity.
17. Which type of MIMO uses multiple antennas at both the transmitter and receiver to improve communication performance?
- A) SIMO (Single Input Multiple Output) B) MISO (Multiple Input Single Output)
C) SISO (Single Input Single Output) D) MIMO (Multiple Input Multiple Output)
18. What is the function of Spatial Multiplexing in MIMO technology?
- A) To reduce the power consumption B) To transmit multiple data streams simultaneously over the same frequency channel
C) To enhance voice call quality D) To limit the number of connected devices
19. What is the role of Beamforming in MIMO technology?
- A) To direct radio signals towards specific users, enhancing signal strength and reducing interference B) To increase network latency
C) To limit the range of communication D) To reduce data rates
20. What is the main purpose of using Carrier Aggregation (CA) in conjunction with MIMO in LTE?
- A) To reduce the number of antennas needed B) To simplify network management
C) To limit the number of connected devices D) To combine multiple frequency bands, increasing data rates and overall capacity

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain how the Capacity scales linearly with the number of added small cells.	4	L*2	1	1,2
b)	Write a program for mm wave, considering the environment of several eNodeB, attaches one UE per eNodeB starts a data flow for each UE to and from a remote host.	8	L3	2	1,2,5 *
c)	Illustrates the main features of the 3GPP standards now in the market, highlighting the trend toward widespread use of spectrum, higher bandwidths, higher spectral efficiency and lower latency.	4	L2	1	1,2
2. a)	Briefly explain the 5G standardization activities in IEEE.	4	L2	2	1,2
b)	Considering the scenario of a eNodeB, attaches to one UE0 and starts moving with velocity of 100 Km/hr towards UE1. Write the program for mobility model of 5G network and Print the new position of UE0.	8	L3	1	1,2,5
c)	Discuss the spectrum access options to address increasing demand for spectrum.	4	L2	2	1,2
3. a)	Elaborate the unique challenges for radio communication at millimeter wave bands.	4	L2	2	1,2
b)	Write a program using the ns-3 simulator considering the scenario one eNodeB, attaches three UEs to the eNodeB, and starts a data flow for each UE to and from a remote host. The program should also instantiate one dedicated bearer per UE and enable bearer deactivation at a specified time.	8	L3	1	1,2,5
c)	Explain the role of antennas at mm wave technology.	4	L2	2	1,2,3
Unit – II					
4. a)	Explain the Software Defined Networking (SDN) for 5 G systems with diagram and highlights its benefits.	8	L2	3	1,2
b)	Elaborate the major features of the 4G LTE D2D concept.	8	L2	4	1,2
5. a)	Explain the relation between functional, logical, orchestration and physical architectures with diagram.	8	L2	3	1,2
b)	Illustrate the performance analysis for 5G D2D RRM concept with suitable examples.	8	L2	4	1,2
6. a)	List out and explain high-level design principles for the 5G architecture.	8	L2	3	1,2
b)	Illustrate the support by the combination and integration of cellular and D2D technologies for National security and public safety requirements with example.	8	L2	4	1,2
Unit – III					
7. a)	Explain the Multiuser and single user with massive MIMO base stations using example.	8	L2	5	1,2
b)	Elaborate the Impact of channel state information errors on the throughput of massive MIMO systems.	8	L2	5	1,2
8. a)	Explain the uplink transmission using coded random access with 5 users u_i and a set of 2 pilot sequences s_1 and s_2 .	8	L2	5	1,2,3
b)	Explain the Decentralized-coordinated transceiver design for massive MIMO.	8	L2	5	1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
